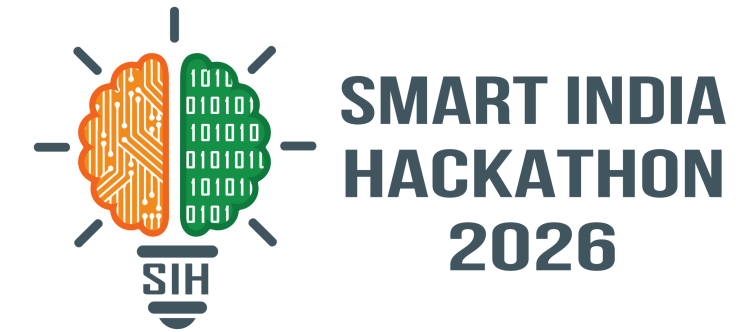


SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26125
- Problem Statement Title- Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team Name - Code geass



PROBLEM

- centralized storage creates a single point of trust and attack.
- Unauthorized alteration is hard to independently verify.
- Sensitive sharing needs controlled access, revocation and auditability.

OUR SOLUTION

- KYC-backed secure document vault for verified users.
- Documents stay encrypted off-chain; blockchain stores proof and access state.
- Wallet signatures authenticate grant, revoke and access actions.

CORE VALUE

Confidentiality
Encrypted Files

Integrity
Blockchain Hash

Control
Grant / Revoke

Accountability
Immutable Audit

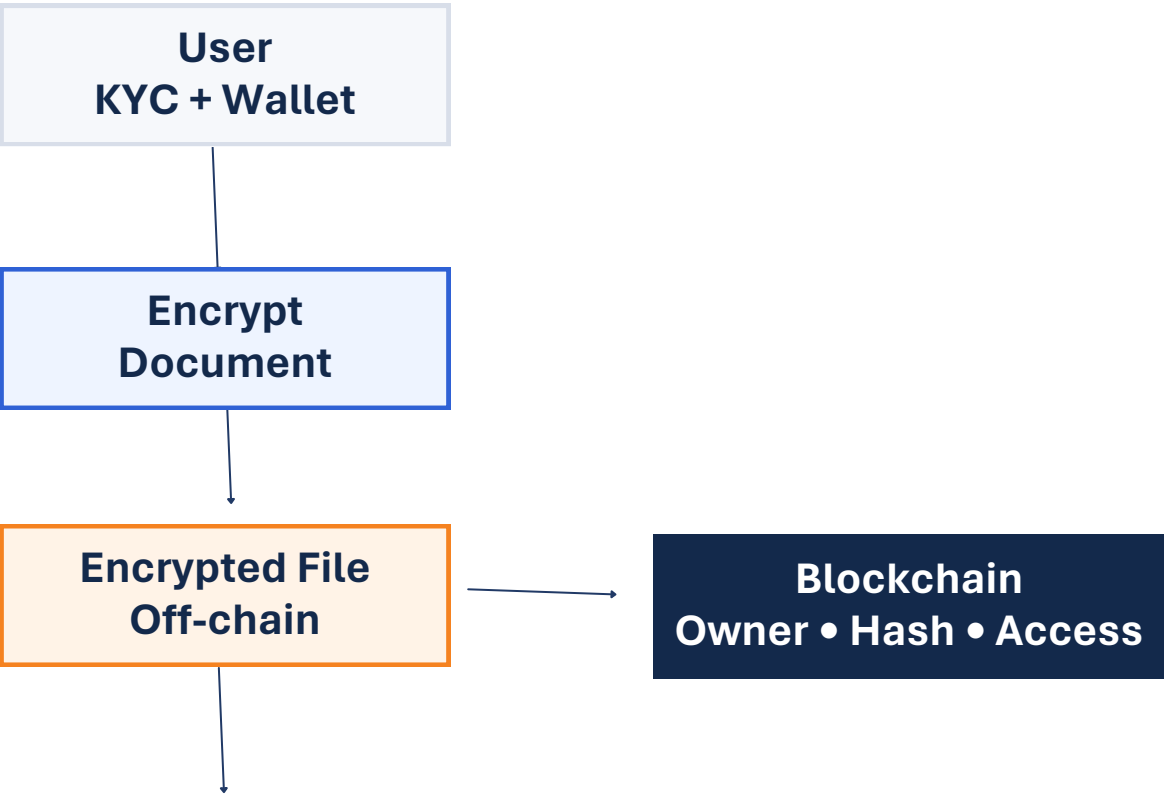
WHY WE'RE DIFFERENT

- Encrypted off-chain storage + on-chain trust layer
- Immutable document integrity through blockchain hash
- KYC-backed wallet identity with grant/revoke control

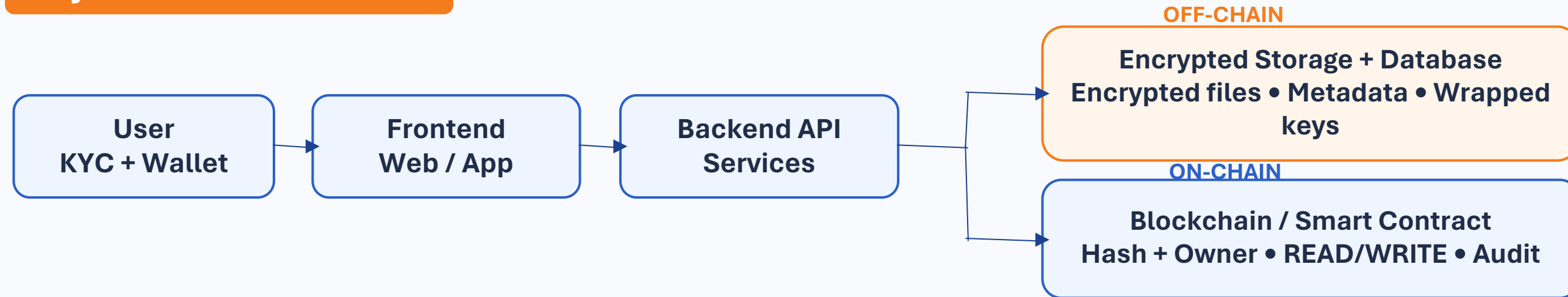
The file is not stored on blockchain.
Blockchain stores the proof layer: ownership, SHA-256 fingerprint, permissions and audit history.

Core idea: use blockchain only where immutability adds value — not as expensive file storage.

Simple Architecture Flow



1. System Architecture



2. Process Flowchart

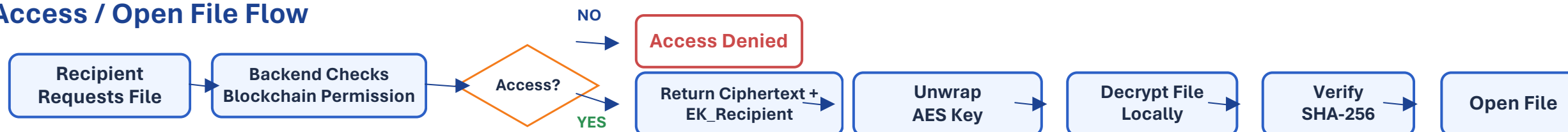
1 Secure Upload Flow



2 Grant Access Flow



3 Access / Open File Flow



4 Revoke Access Flow



Files remain off-chain in encrypted form; blockchain stores proof, permissions and audit state.

TECH / SOFTWARE USED

- HTML
- CSS
- javascript
- MetaMask
- Node js
- ethers js
- express js
- MongoDB
- solidity
- ethereum sepolia
- hardhat
- SHA 256
- AES - 256 - GCM

Technical Feasibility

1) Proven Encryption Stack

AES-256-GCM and asymmetric key wrapping provide established, implementable file-security mechanisms.

2) Compatible Full-Stack Architecture

React, Node.js, MongoDB and Ethers.js integrate smoothly using standard APIs.

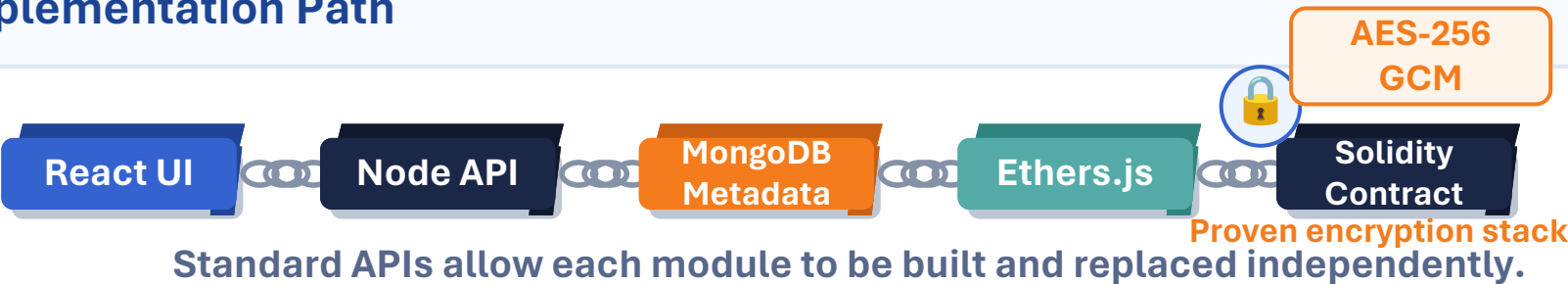
3) Implementable Smart Contracts

Solidity contracts can enforce permissions, revocation, audit events and file-version records.

4) Scalable Hybrid Design

Encrypted files remain off-chain while hashes and access records stay on-chain.

Implementation Path



Economic Approach

1) Cost-Efficient Hybrid Storage

Large files are encrypted and stored off-chain using scalable cloud storage; blockchain stores only lightweight proof data.

2) Lower Verification Overhead

Digitally signed credentials and on-chain records reduce manual verification steps and enable faster credential checks.

3) Reduced Administrative Overhead

Smart contracts record permission changes and predefined access rules, reducing repetitive manual control and audit work.

Cost-Smart Hybrid Model



Viability edge: use blockchain only where immutability adds value — not as expensive file storage.

IMPACTS

National Impact:

Fortifies cyber defenses to prevent data breaches, safeguarding sensitive information and bolstering national security.

Economic Impact:

Reduces the financial overhead associated with centralized data hosting while mitigating the costly risks of ransomware and fraud.

Social Impact:

Empowers citizens by securing their digital identities and fostering systemic trust through a decentralized network.

Impact summary: stronger cyber resilience, lower costs, and higher trust in secure digital interactions.

FUTURE PROSPECTS

Integrating automated smart contracts across enterprise institutions will enable frictionless global credential and identity verification.

Scaling decentralized storage across distributed networks will eliminate centralized database vulnerabilities and single failure points.

Deploying zero knowledge proofs will verify user credentials while preserving absolute confidentiality for sensitive data.

Automating strong revocation workflows will provide immediate key rotation to protect future encrypted file versions.

RESOURCE BASE

SUPPORTING RESEARCH PAPERS

- Gautam & Ali (2025) — Blockchain-based secure document storage and sharing.
DOI: 10.30574/wjaets.2025.15.3.1071
- Gousteris et al. (2023) — Secure distributed cloud storage using blockchain and smart contracts.
DOI: 10.28991/ESJ-2023-07-02-012
- Wang et al. (2019) — Secure cloud storage framework with blockchain-based access control.
DOI: 10.1109/ACCESS.2019.2929205

DATA SOURCES

- Network & Testing: Ethereum-compatible testnets / Ethereum Sepolia and private Hardhat chains.
- Cryptographic Standards: Web Crypto API and libsodium-style AES-GCM encryption + RSA-OAEP key wrapping.
- Smart Contract Vulnerability Registries: best-practice references for issues like re-entrancy and unsafe authorization.

MARKET RESEARCH

- Decentralized Identity market projected at approx. 89% CAGR, reaching over \$46B by 2030.
- Demand is driven by reducing centralized single points of failure, seen in large-scale data leaks such as Equifax and Aadhaar.
- Krypto Vault targets this gap with user-owned keys and fine-grained access revocation.

TECHNICAL DOCUMENTATION

- Databases: MongoDB Atlas, MongoDB GridFS and client-side IndexedDB.
- Frameworks & Tools: HTML, CSS, JavaScript, Node.js, TypeScript, Express, MetaMask and ethers.js.
- Blockchain: Solidity, Hardhat and Ethereum Sepolia.
- Cryptography: AES-256-GCM, SHA-256, RSA-OAEP and PBKDF2-SHA-256.

All project-specific architecture details are available through the extended documentation link / QR.

PROJECT LINKS / QR



Extended Technical Doc

Architecture • Security model
Database • On-chain/off-chain
Revocation / key rotation

<https://drive.google.com/file/d/1BUO6qAf-Oi9J9d91R9oz528d7kw50FT0/view?usp=drivesdk>



Demo Video

Website walkthrough • upload flow
Wallet/KYC screens • backend demo

https://drive.google.com/drive/folders/1sN9ngNXsQU7qOCu-mPmk7jew7MpBMF2FD?usp=drive_link



Source Code / GitHub

Frontend • Backend • Smart contract /
integration code

https://github.com/samarthnockem/sih_project_code_geasst



demo website link

Frontend ui, woking + interface

<https://blockchain-project-sih-1.onrender.com/>